

**6.6 Classwork: The Discriminant and Polynomial Equations (no calculator) —
Markscheme**

1. [Maximum mark: 5]

For each quadratic function, find the discriminant $\Delta = b^2 - 4ac$ and state the number of real roots.

(a) $f(x) = x^2 - 6x + 9$

$$\Delta = (-6)^2 - 4(1)(9) = 36 - 36 = 0 \quad (\text{A1})$$

One repeated real root (the parabola touches the x -axis). (A1)

(b) $g(x) = 2x^2 + 3x + 5$

$$\Delta = 3^2 - 4(2)(5) = 9 - 40 = -31 \quad (\text{A1})$$

No real roots ($\Delta < 0$; the parabola does not cross the x -axis). (A1)

(c) $\Delta = 0$: graph touches the x -axis at one point (repeated root). $\Delta < 0$: graph does not intersect the x -axis. (A1)

2. [Maximum mark: 6]

The equation $x^2 + kx + 9 = 0$ has two equal roots.

(a) $\Delta = k^2 - 4(1)(9) = k^2 - 36 \quad (\text{A1})$

(b) For equal roots, $\Delta = 0$: (M1)

$$k^2 - 36 = 0 \implies k^2 = 36 \implies k = \pm 6 \quad (\text{A1})$$

(c) When $k = 6$: $x^2 + 6x + 9 = (x + 3)^2 = 0$, so $x = -3$. (A1)

When $k = -6$: $x^2 - 6x + 9 = (x - 3)^2 = 0$, so $x = 3$. (A1)

(d) No real roots when $\Delta < 0$: $k^2 - 36 < 0$, i.e. $-6 < k < 6$. (A1)

3. [Maximum mark: 7]

The line $y = 2x + c$ is tangent to the parabola $y = x^2 + 1$.

(a) Setting equal: $x^2 + 1 = 2x + c$

$$x^2 - 2x + 1 - c = 0, \text{ i.e. } x^2 - 2x + (1 - c) = 0 \quad (\text{AG})$$

$$(b) \Delta = (-2)^2 - 4(1)(1 - c) = 4 - 4 + 4c = 4c \quad (\text{M1})$$

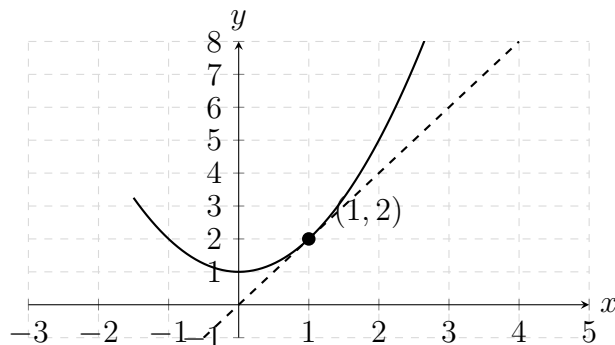
$$\text{For tangency: } \Delta = 0 \implies 4c = 0 \implies c = 0 \quad (\text{A1})$$

$$(c) \text{ With } c = 0: x^2 - 2x + 1 = 0 \implies (x - 1)^2 = 0, \text{ so } x = 1. \quad (\text{M1})$$

$$y = 2(1) = 2. \text{ Point of tangency: } (1, 2). \quad (\text{A1})$$

(d) Sketch: upward parabola through $(0, 1)$ with vertex at $(0, 1)$, and the line $y = 2x$ touching the parabola at $(1, 2)$. (A1)(A1)

Award (A1) for correct parabola shape with y-intercept at 1, (A1) for tangent line touching at $(1, 2)$.



4. [Maximum mark: 6]

The diagram shows the graph of $f(x) = -(x - 3)^2 + 4$.

$$(a) \text{ Vertex: } (3, 4) \quad (\text{A1})$$

$$(b) f(x) = -(x^2 - 6x + 9) + 4 = -x^2 + 6x - 9 + 4 = -x^2 + 6x - 5 \quad (\text{M1})(\text{AG})$$

$$(c) -x^2 + 6x - 5 = 0 \implies x^2 - 6x + 5 = 0 \quad (\text{M1})$$

$$(x - 1)(x - 5) = 0, \text{ so } x = 1 \text{ and } x = 5 \quad (\text{A1})$$

$$(d) \Delta = 6^2 - 4(1)(5) = 36 - 20 = 16 > 0$$

Since $\Delta > 0$, there are two distinct real roots, confirming two x -intercepts. (A1)

5. [Maximum mark: 6]

Consider the function $f(x) = 2x^2 - 8x + 11$.

(a) $f(x) = 2(x^2 - 4x) + 11 = 2(x^2 - 4x + 4 - 4) + 11$ (M1)

$$= 2(x - 2)^2 - 8 + 11 = 2(x - 2)^2 + 3 \quad (\text{A1})$$

(b) Vertex: $(2, 3)$ (A1)

(c) $\Delta = (-8)^2 - 4(2)(11) = 64 - 88 = -24 < 0$ (M1)

Since $\Delta < 0$, the equation has no real solutions. (AG)

(d) Minimum value of $f(x)$ is 3 (the k -value from vertex form). (A1)

6. [Maximum mark: 6]

Let $p(x) = x^3 - 6x^2 + 11x - 6$.

(a) $p(1) = 1 - 6 + 11 - 6 = 0$ (M1)

Since $p(1) = 0$, $(x - 1)$ is a factor by the factor theorem. (R1)

(b) By polynomial division (or inspection):

$$x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6) \quad (\text{M1})$$

$$= (x - 1)(x - 2)(x - 3) \quad (\text{A1})$$

(c) Solutions: $x = 1, x = 2, x = 3$ (A1)

(d) Cubic curve crossing the x -axis at $x = 1, 2, 3$ and the y -axis at $(0, -6)$. Positive leading coefficient gives the standard cubic shape $(-\infty$ to $+\infty)$. (A1)